Phys 115A HW 6

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Question 1. *1 Hermitian Conjugates (Griffiths 3.5 + extra)*
1. Find the Hermitian conjugates of x, i, and d/dx.
2. Show that for any two operators $\hat{Q}$ and $\hat{R}$ (not necessarily Hermitian), that $(\hat{Q}\hat{R})^\dagger = \hat{R}^\dagger\hat{Q}^\dagger$, $(\hat{Q}+\hat{R})^\dagger = \hat{Q}^\dagger + \hat{R}^\dagger$, and $(c\hat{Q})^\dagger = c^*\hat{Q}^\dagger$ for a complex number c.
3. Construct the Hermitian conjugate of the harmonic oscillator raising operator a^+ (Eq. 2.48).
4. For a particle moving in one dimension, show that the operator $\hat{x}\hat{p}$ is not Hermitian. Can you construct an operator corresponding to some sort of physically observable product of position and momentum that is Hermitian?

Proof.

1. Here, if consider the Nuclear Space of integrable functions with domain $\mathbb{R}$, codomain $\mathbb{C}$, then given any such functions ϕ, ψ, we have:

$$\langle\phi|x\psi\rangle = \int_{-\infty}^{\infty} \phi^* \cdot (x\cdot\psi)dx = \int_{-\infty}^{\infty} (x\cdot\phi)^*\cdot\psi dx = \langle x\phi|\psi\rangle \tag{1}$$

$$\langle\phi|i\psi\rangle = \int_{-\infty}^{\infty} \phi^* \cdot (i\psi)dx = \int_{-\infty}^{\infty} (-i\phi)^*\cdot\psi dx = \langle -i\phi|\psi\rangle \tag{2}$$

$$\left\langle\phi\middle|\frac{d}{dx}\psi\right\rangle = \int_{-\infty}^{\infty} \phi^*\cdot\frac{d}{dx}\psi dx = \phi^*\psi\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\left(\frac{d}{dx}\phi^*\right)\cdot\psi dx \tag{3}$$

$$= \int_{-\infty}^{\infty}\left(-\frac{d}{dx}\phi\right)^*\cdot\psi dx = \left\langle -\frac{d}{dx}\phi\middle|\psi\right\rangle \tag{4}$$

So, the above three relations show that $x^\dagger = x$, $i^\dagger = -i$, and $\left(\frac{d}{dx}\right)^\dagger = -\frac{d}{dx}$.

2. Given any two operators $\hat{Q}, \hat{R}$, let $|\phi\rangle, |\psi\rangle$ be any two vectors, then:

$$\left\langle(\hat{Q}\hat{R})^\dagger\phi\middle|\psi\right\rangle = \left\langle\phi\middle|(\hat{Q}\hat{R})\psi\right\rangle = \left\langle\hat{Q}^\dagger\phi\middle|\hat{R}\psi\right\rangle = \left\langle\hat{R}^\dagger\hat{Q}^\dagger\phi\middle|\psi\right\rangle \tag{5}$$

$$\left\langle(\hat{Q}+\hat{R})^\dagger\phi\middle|\psi\right\rangle = \left\langle\phi\middle|(\hat{Q}+\hat{R})\psi\right\rangle = \left\langle\phi\middle|\hat{Q}\psi\right\rangle + \left\langle\phi\middle|\hat{R}\psi\right\rangle \tag{6}$$

$$= \left\langle\hat{Q}^\dagger\phi\middle|\psi\right\rangle + \left\langle\hat{R}^\dagger\phi\middle|\psi\right\rangle = \left\langle(\hat{Q}^\dagger+\hat{R}^\dagger)\phi\middle|\psi\right\rangle \tag{7}$$

$$\left\langle (c\hat{Q})^\dagger \phi \middle| \psi \right\rangle = \left\langle \phi \middle| c\hat{Q}\psi \right\rangle = c\left\langle \phi \middle| \hat{Q}\psi \right\rangle = c\left\langle \hat{Q}^\dagger \phi \middle| \psi \right\rangle = \left\langle c^* \hat{Q}^\dagger \phi \middle| \psi \right\rangle \tag{8}$$

Which, the above three relations show that $(\hat{Q}\hat{R})^\dagger = \hat{R}^\dagger \hat{Q}^\dagger$, $(\hat{Q} + \hat{R})^\dagger = \hat{Q}^\dagger + \hat{R}^\dagger$, and $(c\hat{Q})^\dagger = c^* \hat{Q}^\dagger$.

3. Recall that the raising operator is defined as $a_+ = \frac{1}{\sqrt{2m\hbar w}}\left(x - i\frac{d}{dx}\right)$. So, using the result in part 1 and 2, we have:

$$a_+^\dagger = \left(\frac{1}{\sqrt{2m\hbar w}}\left(x - i\frac{d}{dx}\right)\right)^\dagger = \frac{1}{\sqrt{2m\hbar w}}\left(x^\dagger - \left(i\frac{d}{dx}\right)^\dagger\right) = \frac{1}{\sqrt{2m\hbar w}}\left(x + i\frac{d}{dx}\right) = a_- \tag{9}$$

Hence, the adjoint (Hermitian conjugate) of the raising operator is the lowering operator.

4. Under one dimension, one has $\hat{x}\psi = x\psi$, and $\hat{p}\psi = -i\hbar\frac{d}{dx}\psi$. And, recall that both are self-adjoint/Hermitian operators (on the function space). So, we have the following:

$$(\hat{x}\hat{p})^\dagger \psi = \hat{p}\hat{x}\psi = -i\hbar\frac{d}{dx}(x\psi) = -i\hbar\psi + x \cdot \left(-i\hbar\frac{d}{dx}\psi\right) = \left(-i\hbar + \hat{x}\hat{p}\right)\psi \tag{10}$$

So, $(\hat{x}\hat{p})^\dagger = \hat{x}\hat{p} - i\hbar \neq \hat{x}\hat{p}$, showing it's not self-adjoint/Hermitian. However, if consider the operator $\frac{\hat{x}\hat{p}+\hat{p}\hat{x}}{2}$, since $(\hat{x}\hat{p})^\dagger = \hat{p}\hat{x}$ and $(\hat{p}\hat{x})^\dagger = \hat{x}\hat{p}$, we have:

$$\left(\frac{\hat{x}\hat{p} + \hat{p}\hat{x}}{2}\right)^\dagger = \frac{(\hat{x}\hat{p})^\dagger + (\hat{p}\hat{x})^\dagger}{2} = \frac{\hat{p}\hat{x} + \hat{x}\hat{p}}{2} = \frac{\hat{x}\hat{p} + \hat{p}\hat{x}}{2} \tag{11}$$

Which is a desired self-adjoint/Hermitian operator corresponding to somewhat a product of position and momentum.

□

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Question 2. *2 Hilbert Space*
1. Show that the set of all square-integrable functions (that is, the set of functions $f(x)$ for which $\int_a^b |f(x)|^2 dx$ is finite on a specified interval $x \in [a,b]$) is a vector space.
2. Show that $\langle f|g\rangle = \int_a^b f^*(x)g(x)dx$ satisfies the conditions for an inner product.

Proof.

1. Let L denotes the set of all square-integrable functions on $[a,b]$. Then, for any $f,g \in L$ and any $z \in \mathbb{C}$, we have:

$$|f(x)+g(x)|^2 \leq (|f(x)|+|g(x)|)^2 \leq (2\max\{|f(x)|,|g(x)|\})^2 = 4\max\{|f(x)|^2,|g(x)|^2\} \leq 4(|f(x)|^2+|g(x)|^2) \tag{12}$$

Hence, performing integration, one gets:

$$\int_a^b |f(x)+g(x)|^2 ds \leq 4\left(\int_a^b |f(x)|^2 dx + \int_a^b |g(x)|^2 dx\right) < \infty \tag{13}$$

Hence, $f(x)+g(x) \in L$, which L is closed under addtion. Similarly, one has:

$$\int_a^b |zf(x)|^2 dx = |z|^2 \int_a^b |f(x)|^2 dx < \infty \tag{14}$$

Hence, $zf(x) \in L$, showing L is closed under $\mathbb{C}$-scalar multiplication.

So, these two conditions conclude that L is a $\mathbb{C}$-vector space.

2. To show the integral is an inner product, given any $f,g,h \in L$ and any $z,w \in \mathbb{C}$, we have:

$$\langle f|f\rangle = \int_a^b f(x)^* f(x)dx = \int_a^b |f(x)|^2 dx \geq 0 \tag{15}$$

$$\langle f|f\rangle = 0 \iff \int_a^b |f(x)|^2 dx = 0 \iff |f(x)|^2 = 0 \iff f(x) = 0 \tag{16}$$

(Note: for this one, it requires the continuity of $|f(x)|^2$ on $[a,b]$. For non-continuous functions, there ar pathological cases where $f(x) \neq 0$, yet such integral is 0. For instance, $f(a) = 1$ and $f(x) = 0$ elsewhere works. In general, if the set of values where $f(x) \neq 0$ is measure 0, then such integral is 0).

$$\langle f|zg+wh\rangle = \int_a^b f(x)^*(zg(x)+wh(x))dx = z\int_a^b f(x)^* g(x)dx + w\int_a^b f(x)^* h(x)dx = z\langle f|g\rangle + w\langle f|h\rangle \tag{17}$$

$$\langle g|f\rangle = \int_a^b g(x)^* f(x)dx = \int_a^b (f(x)^* g(x))^* dx = \left(\int_a^b f(x)^* g(x)dx\right)^* = \langle f|g\rangle^* \tag{18}$$

So, these conditions characterize an inner product on L.

$\square$

3 (part 3 a, c ND)

Question 3. *3 Rigged Hilbert Space*
1. Consider a Hilbert space H that consists of all functions $\psi(x)$ such that $\int_{-\infty}^{\infty} |\psi(x)|^2 dx$ is finite. Show that there are functions in H for which $\hat{x}\psi(x) \equiv x\psi(x)$ is not in H.
2. Consider the function space Ω, called the nuclear space, which consists of all functions $\phi(x)$ that satisfy the infinite set of conditions $\int_{-\infty}^{\infty} |\phi(x)|^2(1+|x|)^n dx < \infty$ for $n = 0, 1, 2, \ldots$. Show that for any $\phi(x)$ in Ω, the function $\hat{x}\phi(x) \equiv x\phi(x)$ is also in Ω.
3. Now consider the extended space $\Omega^\times$, which consists of all functions $\chi(x)$ that satisfy the condition $\langle\chi|\phi\rangle = \int_{-\infty}^{\infty} \chi^*(x)\phi(x)dx < \infty$ for all ϕ in the nuclear space Ω. Which of the following functions belong to Ω, to H, and/or to $\Omega^\times$?
(a) $\sin(x)$ *(b)* $\sin(x)/x$ *(c)* $x^2\cos(x)$ *(d)* $\dfrac{\log(1+|x|)}{1+|x|}$ *(e)* $\exp(-x^2)$ *(f)* $x^4 e^{-|x|}$

Proof.

1. Consider the function $\psi(x) = \frac{1}{\sqrt{x^2+1}}$, then $|\psi(x)|^2 = \frac{1}{x^2+1}$, showing $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \arctan(x)\Big|_{-\infty}^{\infty} = \pi$, hence $\psi(x) \in H$. However, if consider $x\psi(x)$, one has $|x\psi(x)|^2 = \frac{x^2}{x^2+1} = 1 - \frac{1}{x^2+1}$, which the integral $\int_a^b \left(1 - \frac{1}{x^2+1}\right) dx = (b-a) - \left(\arctan(x)\Big|_a^b\right) = (b-a) - (\arctan(b) - \arctan(a))$. Which, take $b \to \infty$ and $a \to -\infty$, this integral clearly diverges (since $\arctan(b) - \arctan(a)$ is bounded, but not so for $b-a$). So, $x\psi(x) \notin H$.

2. First, by induction one can show that $\int_{-\infty}^{\infty} |\phi(x)|^2 \cdot |x|^n dx < \infty$ for all $n \in \mathbb{N}$. For base case $n = 0$, it is given by $\int_{-\infty}^{\infty} |\phi(x)|^2 \cdot |x|^0 dx = \int_{-\infty}^{\infty} |\phi(x)|^2 \cdot (1+|x|)^0 dx < \infty$. Now, suppose for all integer $k < n$ it satisfies $\int_{-\infty}^{\infty} |\phi(x)|^2 \cdot |x|^k dx < \infty$, then for $k = n$, one has the following:
$$|x|^n = (1+|x|)^n - \sum_{k=0}^{n-1} \binom{n}{k} |x|^k \tag{19}$$
$$\implies \int_{-\infty}^{\infty} |\phi(x)|^2 \cdot |x|^n dx = \int_{-\infty}^{\infty} |\phi(x)|^2 \cdot (1+|x|)^n dx - \sum_{k=0}^{n-1} \binom{n}{k} \int_{-\infty}^{\infty} |\phi(x)|^2 \cdot |x|^k dx < \infty \tag{20}$$
So, $\int_{-\infty}^{\infty} |\phi(x)|^2 \cdot |x|^n dx < \infty$, which completes the induction.
So, this implies that given $\phi(x) \in \Sigma$, one has $x\phi(x)$ satisfies the following for all $n \in \mathbb{N}$:
$$\int_{-\infty}^{\infty} |x\phi(x)|^2(1+|x|)^n dx = \sum_{k=0}^{n} \binom{n}{k} \int_{-\infty}^{\infty} |\phi(x)|^2 \cdot |x|^{k+2} dx < \infty \tag{21}$$
So, this indicates that $x\phi(x) \in \Omega^\times$ also.

3. The spaces satisfy $\Omega \subseteq H \subseteq \Omega^\times$. Which, the functions given satisfy:
   (a) $\sin(x) \in \Omega^\times$: Given that $\int_{-\infty}^{\infty} |\sin(x)|^2 dx$ diverges (since integral of $\sin^2(x)$ diverges), then $\sin(x) \notin H$; however, since $-1 \le \sin(x) \le 1$, any $\phi(x) \in \Omega^\times$ satisfies

(b) $\frac{\sin(x)}{x} \in H \setminus \Omega$, since one has $\int_{-\infty}^{\infty} \left|\frac{\sin(x)}{x}\right|^2 dx < \infty$ (since around $x = 0$, one has $\int_{-1}^{1} \frac{\sin^2(x)}{x^2} dx$ converges due to the existence of limit at $x = 0$, and the other two pieces $\int_1^{\infty} \frac{\sin^2(x)}{x^2} dx \leq \int_1^{\infty} \frac{1}{x^2} dx < \infty$, similar for the other piece on $(-\infty, -1)$).

However, it's not in Ω, since $x \cdot \frac{\sin(x)}{x} = \sin(x) \notin H$ (by part 3 (a)), hence $x \cdot \frac{\sin(x)}{x} \notin \Omega$, which it can't be in Ω (or else it contradicts the result in part 2).

(c) d

(d) $\frac{\log(1+|x|)}{1+|x|} \in H \setminus \Omega$. Recall that for all $\epsilon > 0$, the growth order of $\log(z)$ is less than z^{ϵ}. So, for simplicity one can pick $\epsilon = \frac{1}{4}$, so there exists $a \in \mathbb{R}$, such that $\log(z) < z^{\frac{1}{4}}$ for all $a \in \mathbb{R}$. As a result, one has the following:

$$\int_a^{\infty} \left|\frac{\log(1+|x|)}{1+|x|}\right|^2 dx < \int_a^{\infty} \frac{(1+|x|)^{\frac{1}{2}}}{(1+|x|)^2} dx = \int_a^{\infty} \frac{1}{(1+|x|)^{\frac{3}{2}}} dx < \infty \tag{22}$$

Which, $\int_0^{\infty} \left|\frac{\log(1+|x|)}{1+|x|}\right|^2 dx < \infty$, with the function being symmetric, this implies $\int_{-\infty}^{\infty} \left|\frac{\log(1+|x|)}{1+|x|}\right|^2 dx < \infty$.

The reason it's not in Ω, because take $n = 2$, one has the following:

$$\int_{-\infty}^{\infty} \left|\frac{\log(1+|x|)}{1+|x|}\right|^2 \cdot (1+|x|)^2 dx = \int_{-\infty}^{\infty} \log^2(1+|x|) dx \to \infty \tag{23}$$

Which, it doesn't satisfy the condition for being in Ω.

(e) $e^{-x^2} \in \Omega$, since for any $n \in \mathbb{N}$, $(1+|x|)^n$ has growth order being a polynomial, so one has the following:

$$\int_{-\infty}^{\infty} |e^{-x^2}|^2 (1+|x|)^n dx = \int_{-\infty}^{\infty} e^{-2x^2} (1+|x|)^n dx < \infty \tag{24}$$

Since e^{-2x^2} has growth order much larger than the polynomials, the above integral converges.

(f) $x^4 e^{-|x|} \in \Omega$, since for any $n \in \mathbb{N}$, one has $(1+|x|)^n$ being a polynomial, so is $|x|^8 (1+|x|)^n$. As a result, one has:

$$\int_{-\infty}^{\infty} |x^4 e^{-|x|}|^2 (1+|x|)^n dx = \int_{-\infty}^{\infty} e^{-2|x|} \cdot |x|^8 (1+|x|)^n dx < \infty \tag{25}$$

Since the improper integral on both sides are with exponential functions, which decay at a faster rate than any polynomials.

$\square$

4 (part 6 ND)

Question 4. *4 Operators, Eigenvectors, Eigenvalues*
What are the eigenvectors and eigenvalues for the following operators? In each case, how would you express a general initial wavefunction $\Psi(x,0)$ in terms of linear combinations of these eigenvectors?
1. The position operator $\hat{x}$ in position space.
2. The momentum operator $\hat{p}$ in position space.
3. The Hamiltonian for the free particle in position space.
4. The Hamiltonian for the infinite square well in position space.
5. The Hamiltonian for the harmonic oscillator in position space.
6. The lowering operator a^- for the harmonic oscillator in position space.
7. The operator $i\frac{d}{d\phi}$, where ϕ is the polar angle in two dimensions, acting on the space of functions defined on the interval $0 \leq \phi \leq 2\pi$ such that $f(\phi + 2\pi) = f(\phi)$.
8. The operator $\frac{d^2}{d\phi^2}$, acting on the same space.

Proof.

1. Given $\hat{x} = x$, the eigenvectors are $\delta(x-a)$, corresponding to eigenvalue $a \in \mathbb{R}$ (since $x\delta(x-a) = a\delta(x-a)$ as operators when integrating with general functions). For general initial wavefunction $\Psi(x,0)$, one can express it as:

$$\Psi(x,0) = \int_{-\infty}^{\infty} \Psi(a,0)\delta(x-a)da \tag{26}$$

2. Given $\hat{p} = -i\hbar\frac{d}{dx}$, the eigenvectors and eigenvalues are $\frac{1}{\sqrt{2\pi}}e^{ikx}$, with eigenvalue $\hbar k$ for $k \in \mathbb{R}$ (since taking derivative one yields $\frac{ik}{\sqrt{2\pi}}e^{ikx}$, multiply $-i\hbar$ gives $\frac{\hbar k}{\sqrt{2\pi}}e^{ikx}$). For general initial wavefunction $\Psi(x,0)$, one can express it as (using Fourier / Inverse Fourier Transform):

$$\Psi(x,0) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \phi(k)e^{ikx}dk, \quad \phi(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \Psi(x,0)e^{-ikx}dx \tag{27}$$

3. Given Hamiltonia for the free partical $\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$, the eigenvectors are again $\frac{1}{\sqrt{2\pi}}e^{ikx}$ for $k \in \mathbb{R}$, with eigenvalue $\frac{k^2\hbar^2}{2m}$ (since after taking second derivative, one yields $-\frac{k^2}{\sqrt{2\pi}}e^{ikx}$, multiply by $-\frac{\hbar^2}{2m}$ gives $\frac{k^2\hbar^2}{2m} \cdot \frac{1}{\sqrt{2\pi}}e^{ikx}$). For general initial wavefunction $\Psi(x,0)$, one again express it as follow, similar to part 2:

$$\Psi(x,0) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \phi(k)e^{ikx}dk, \quad \phi(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \Psi(x,0)e^{-ikx}dx \tag{28}$$

4. The Hamiltonian for the infinite square well is $\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$, where $x \in [0,a]$, with condition $\psi(0) = \psi(a) = 0$, and $\psi(x) = 0$ for $x \notin [0,a]$. The eigenvectors are $\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)$ ($x \in [0,a]$) for $n \in \mathbb{Z}^{\geq 0}$, with eigenvalue $E_n = \frac{1}{2m}\left(\frac{n\pi\hbar}{a}\right)^2$ (since after taking second derivative, one yields $-\frac{n^2\pi^2}{a^2} \cdot \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)$,

multiply by $-\frac{\hbar^2}{2m}$ gives $\frac{n^2\pi^2\hbar^2}{2ma^2}\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)$. For general initial wavefunction $\Psi(x,0)$ can be expressed as follow:

$$\Psi(x,0)=\sum_{n=1}^{\infty}c_n\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right),\quad c_n=\int_0^a\Psi(x,0)\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)dx \tag{29}$$

5. The Hamiltonian for harmonic oscillator is $\hat{H}=-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+\frac{1}{2}mw^2x^2$, so the eigenvectors are given by $\psi_n(x)=\frac{1}{\sqrt{n!}}(a_+)^n\psi_0(x)$, where $\psi_0(x)=\left(\frac{mw}{\pi\hbar}\right)^{\frac{1}{4}}e^{-\frac{mw}{2\hbar}x^2}$ and the operator $a_+=\frac{1}{\sqrt{2mw\hbar}}(\hat{x}-i\hat{p})$, with eigenvalue $E_n=\hbar w\left(n+\frac{1}{2}\right)$ (these are the results from the Harmonic Oscillator chapter). For general initial wavefunction $\Psi(x,0)$, one can express it as:

$$\Psi(x,0)=\sum_{n=0}^{\infty}c_n\psi_n(x),\quad c_n=\int_{-\infty}^{\infty}\Psi(x,0)\psi_n(x)dx \tag{30}$$

6. Given the lowering operator $a_-=\frac{1}{\sqrt{2mw\hbar}}(mw\hat{x}+i\hat{p})=\frac{1}{\sqrt{2mw\hbar}}\left(mwx+\hbar\frac{d}{dx}\right)$. Which, suppose $\lambda\in\mathbb{C}$ and $\psi:\mathbb{R}\to\mathbb{C}$ satisfies the following:

$$a_-\psi=\lambda\psi\implies mwx\psi+\hbar\frac{d\psi}{dx}=\sqrt{2mw\hbar}\lambda\psi \tag{31}$$

$$\implies\frac{d\psi}{dx}+\left(\frac{mw}{\hbar}x-\sqrt{\frac{2mw}{\hbar}}\lambda\right)\psi=0 \tag{32}$$

Which, using integrating factor, we get:

$$I=\exp\left(\int\left(\frac{mw}{\hbar}x-\sqrt{\frac{2mw}{\hbar}}\lambda\right)dx\right)=\exp\left(\frac{mw}{2\hbar}x^2-\sqrt{\frac{2mw}{\hbar}}\lambda x\right) \tag{33}$$

$$\implies I\psi=C\implies\psi=CI^{-1}=C\exp\left(-\frac{mw}{2\hbar}x^2+\sqrt{\frac{2mw}{\hbar}}\lambda x\right) \tag{34}$$

Which, take $ik:=\sqrt{\frac{2mw}{\hbar}}\lambda$,

7. Given $f(\phi+2\pi)=f(\phi)$, if it's the eigenvector of $i\frac{d}{d\phi}$ with eigenvalue λ, one has $i\frac{d}{d\phi}f=\lambda f$, or $\frac{df}{d\phi}=-i\lambda f$, showing $f(\phi)=Ke^{-i\lambda\phi}$. As a result, with period being multiples of 2π, it enforces $\lambda=n$ for $n\in\mathbb{Z}$. So, the eigenvectors can be chosen as $\frac{1}{\sqrt{2\pi}}e^{in\phi}$ for $n\in\mathbb{Z}$, with eigenvalue $-n$. For general initial wavefunction $\Psi(\phi,0)$, one can express it as follow (using Fourier series):

$$\Psi(x,0)=\sum_{n\in\mathbb{Z}}\frac{c_n}{\sqrt{2\pi}}e^{in\phi},\quad c_n=\int_0^{2\pi}\Psi(\phi,0)\frac{1}{\sqrt{2\pi}}e^{-in\phi}d\phi \tag{35}$$

8. Given f as an eigenvector of $\frac{d^2}{d\phi^2}$ with eigenvalue $-\lambda^2$, one has $\frac{d^2f}{d\phi^2}=-\lambda^2\phi$, showing $f(\phi)=Ke^{i\lambda\phi}+Le^{-i\lambda\phi}$. Again, with period being multiples of 2π, this again enforces $\lambda=n$ for $n\in\mathbb{Z}$. So, the eigenvectors can be collected to $\frac{1}{\sqrt{2\pi}}e^{in\phi}$ again, with eigenvalue $-n^2$. For general inital wavefunction $\Psi(\phi,0)$, one can again express it identically as part 7:

$$\Psi(x,0)=\sum_{n\in\mathbb{Z}}\frac{c_n}{\sqrt{2\pi}}e^{in\phi},\quad c_n=\int_0^{2\pi}\Psi(\phi,0)\frac{1}{\sqrt{2\pi}}e^{-in\phi}d\phi \tag{36}$$

□

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Question 5. *5 (Your Name) Uncertainty Principle (Griffiths 3.15)*
Using the generalized uncertainty principle, prove the "(Your Name) Uncertainty Principle" relating the uncertainty in position σ_x to the uncertainty in energy σ_H:

$$\sigma_x \sigma_H \geq \frac{\hbar}{2m} |\langle p \rangle|$$

for $H = \hat{p}^2/2m + V(\hat{x})$. For stationary states this is not particularly helpful – why not?

Proof. Given any two operators $\hat{A}, \hat{B}$ corresponding to observables A, B, the generalized uncertainty principle states:

$$\sigma_A \sigma_B \geq \frac{1}{2} \left| \left\langle [\hat{A}, \hat{B}] \right\rangle \right| \tag{37}$$

Here, choose $A = x$ and $B = H$, then $[\hat{x}, \hat{H}]$ can be given as follow:

$$\begin{aligned}
[\hat{x}, \hat{H}] f &= x \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) f - \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right) (xf) && (38) \\
&= -x \cdot \frac{\hbar^2}{2m} \frac{d^2 f}{dx^2} + xV(x) f + \frac{\hbar^2}{2m} \frac{d^2}{dx^2} (xf) - xV(x) f && (39) \\
&= -\frac{\hbar^2}{2m} \cdot x \frac{d^2 f}{dx^2} + \frac{\hbar^2}{2m} \left(2 \frac{df}{dx} + x \frac{d^2 f}{dx^2} \right) && (40) \\
&= \frac{i\hbar}{m} \left(-i\hbar \frac{d}{dx} \right) f = -\frac{i\hbar}{m} \hat{p} f && (41)
\end{aligned}$$

So, it's concluded that on the function space, $[\hat{x}, \hat{H}] = -\frac{i\hbar}{m}\hat{p}$. Hence, applying the generalized uncertainty principle, one has:

$$\sigma_x \sigma_H \geq \frac{1}{2} \left| \left\langle [\hat{x}, \hat{H}] \right\rangle \right| = \frac{1}{2} \left| \left\langle -\frac{i\hbar}{m} \hat{p} \right\rangle \right| = \frac{\hbar}{2m} \left| \langle \hat{p} \rangle \right| \tag{42}$$

For stationary state, this is not usefule since $\sigma_H = 0$ (as it is an eigenvector of $\hat{H}$), so the inequality is enforces to be $0 \geq \frac{\hbar}{2m} |\langle p \rangle|$, or enforcing $\langle p \rangle = 0$. This doesn't provide further information about the state's uncertainty in position. □

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Question 6. *6 Spectra of Operators (Griffiths 3.16 + extra)*
1. Show that two noncommuting operators cannot have a complete set of common eigenfunctions.
2. Consider two operators $\hat{P}, \hat{Q}$ for which $[\hat{P}, \hat{Q}] = 0$. $\hat{P}$ has a spectrum of non-degenerate eigenvectors ψ_i with eigenvalues p_i. You prepare a particle. A measurement of P yields value p_5. If you measure Q afterwards, what do you find?
3. $\hat{Q}$ has only non-degenerate eigenvectors ψ_i and eigenvalues Q_i. What are the eigenvectors and eigenvalues of the inverse operator $\hat{Q}^{-1}$?

Proof.

1. We'll prove the contrapositive: Suppose $\hat{P}, \hat{Q}$ are two operators with complete set of common eigenvectors $\{\psi_i\}_{i\in I}$, each associated with eigenvalue $\{p_i\}_{i\in I}$ for $\hat{P}$ and $\{q_i\}_{i\in I}$ for $\hat{Q}$ (or, the eigenvectors span the whole space). Then, for any function f, one can express it as $f = \sum_{i\in I} a_i\psi_i$, where $a_i \in \mathbb{C}$ are chosen so that this converges. So, consider the action of $[\hat{P}, \hat{Q}]$, we have:
$$[\hat{P}, \hat{Q}]f = \sum_{i\in I} a_i(\hat{P}\hat{Q} - \hat{Q}\hat{P})\psi_i = \sum_{i\in I} a_i(p_iq_i - q_ip_i)\psi_i = 0 \tag{43}$$
(Note: since $\hat{Q}\psi_i = q_i\psi_i$, and $\hat{P} = p_i\psi_i$).
Hence, on the desired vector space, $[\hat{P}, \hat{Q}] = 0$, showing $\hat{P}, \hat{Q}$ commutes.

2. Since $[\hat{P}, \hat{Q}] = 0$, one has $\hat{P}\hat{Q} = \hat{Q}\hat{P}$. Which, applying on the eigenvectors ψ_i, one has:
$$\hat{P}(\hat{Q}\psi_i) = \hat{Q}(\hat{P}\psi_i) = p_i\hat{Q}\psi_i \tag{44}$$
Which, let $E(\hat{P}, p_i)$ denotes the eigenspace of $\hat{P}$ with eigenvalue p_1, then $\hat{Q}\psi_i \in E(\hat{P}, p_i)$ (showing $\hat{Q}$ maps the eigenspace to itself).
Hence, if measurement of P yields value p_5, the state is currently at ψ_5; which, after measuring Q, one has $\hat{Q}\psi_t \in E(\hat{P}, p_5)$, showing that if measuring P again, one definitively gets p_5. (Even though the state may change, but it's still within the sam eigenspace of $\hat{P}$, so the measured observable of P stays the same).

3. The eigenvectors and eigenvalues of $\hat{Q}^{-1}$ are precisely given by ψ_i and $\frac{1}{Q_i}$ respectively.
First, the ψ_i, $\frac{1}{Q_i}$ are eigenvectors / eigenvalues of $\hat{Q}^{-1}$, since:
$$\psi_i = \hat{Q}^{-1}(\hat{Q}\psi_i) = \hat{Q}^{-1}(Q_i\psi_i) \implies \hat{Q}^{-1}\psi_i = \frac{1}{Q_i}\psi_i \tag{45}$$
Conversely, if v, $\frac{1}{\lambda}$ are eigenvector / eigenvalue of $\hat{Q}^{-1}$, then:
$$v = \hat{Q}(\hat{Q}^{-1}v) = \hat{Q}\left(\frac{1}{\lambda}v\right) \implies \hat{Q}v = \lambda v \tag{46}$$
S o, v, λ are eigenvector / eigenvalue of $\hat{Q}$, by the claim one must have $v = \psi_i$ and $\lambda = Q_i$ for some i. So, the eigenvector / eigenvalue of $\hat{Q}^{-1}$ must be ψ_i and $\frac{1}{Q_i}$ respectively.

□

7

Question 7. *7 Sequential Measurements (Griffiths 3.33)*
Consider an observable A with eigenvectors ψ_1, ψ_2 and eigenvalues a_1, a_2. Consider a second observable B with eigenvectors ϕ_1, ϕ_2 and eigenvalues b_1, b_2. The TA tells you:

$$\psi_1 = \frac{1}{5}(3\phi_1 + 4\phi_2), \qquad \psi_2 = \frac{1}{5}(4\phi_1 - 3\phi_2)$$

1. *You prepare a particle and measure A, getting a_1. What is the state afterwards?*
2. *Then measure B. What are the results and their probabilities?*
3. *Immediately measure A again. What is the probability of getting a_1?*

Proof.

1. If measure observable A and get a_1, it implies the state collapses to the eigenvector of A corresponding to a_1, which is ψ_1.

2. Since after measuring A, the current state is $\psi_1 = \frac{1}{5}(3\phi_1 + 4\phi_2)$ (in terms of the eigenvectors of B), so measuring B, the possible results are the eigenvalues of B, which are b_1, b_2.

   And, since in this state ϕ_1 (the eigenvector of b_1) has coefficient $\frac{3}{5}$, while ϕ_2 (the eigenvector of b_2) has coefficient $\frac{4}{5}$, the probabilities of getting b_1, b_2 are $\mathbb{P}(b_1) = \left|\frac{3}{5}\right|^2 = \frac{9}{25} = 36\%$, and $\mathbb{P}(b_2) = \left|\frac{4}{5}\right|^2 = \frac{16}{25} = 64\%$.

3. First, we'll express ϕ_1, ϕ_2 (the eigenvectors of B, corresponding to eigenvalues b_1, b_2) in terms of ψ_1, ψ_2 (the eigenvectors of A). Which, based on the given relation, by scalar multiplication we get:

$$20\psi_1 = 12\phi_1 + 16\phi_2, \quad 15\psi_2 = 12\phi_1 - 9\phi_2 \tag{47}$$

$$\implies 20\psi_1 - 15\psi_2 = 25\phi_2, \quad \phi_2 = \frac{1}{5}(4\psi_1 - 3\psi_2) \tag{48}$$

$$\tag{49}$$

   Again, with scalar multiplication and substitution, we have:

$$25\psi_1 = 15\phi_1 + 20\phi_2 = 15\phi_1 + 16\psi_1 - 12\psi_2 \tag{50}$$

$$\implies 15\phi_1 = 9\psi_1 + 12\psi_2, \quad \phi_1 = \frac{1}{5}(3\psi_1 + 4\psi_2) \tag{51}$$

   Since it's possible that the state becomes ϕ_1 (with probability 36%) or ϕ_2 (with probability 64%). After measuring A, the probability of getting a_1 is given by the conditional probability.

   If ending with b_1 previously (or at state ϕ_1), since ψ_1 (the eigenvector of a_1) has ϕ_1 coefficient $\frac{3}{5}$, the probability of getting a_1 is $\mathbb{P}(a_1|b_1) = \left|\frac{3}{5}\right|^2 = \frac{9}{25} = 36\%$; else, if ending with b_2 previously (or at state ϕ_2), since ψ_1 has ϕ_1 coefficient $\frac{4}{5}$, the probability of getting a_1 is $\mathbb{P}(a_1|b_2)\left|\frac{4}{5}\right|^2 = 64\%$. So, the total probability of measuring a_1 is:

$$\mathbb{P}(a_1) = \mathbb{P}(a_1|b_1) \cdot \mathbb{P}(b_1) + \mathbb{P}(a_1|b_2) \cdot \mathbb{P}(b_2) = \left(\frac{9}{25}\right)^2 + \left(\frac{16}{25}\right)^2 = \frac{337}{625} \approx 53.92\% \tag{52}$$

$\square$

8

Question 8. *8 Harmonic Oscillator in "energy space" (Griffiths 3.39)*

1. *Compute* $\langle n|\hat{x}|n'\rangle$ *for general* n, n'.
2. *Compute* $\langle n|\hat{p}|n'\rangle$.
3. *Construct the infinite-dimensional matrix representation of* $\hat{x}$ *using these matrix elements.*
4. *Construct the infinite-dimensional matrix for* $\hat{p}$.
5. *Show that* $H = \frac{1}{2m}p^2 + \frac{1}{2}m\omega^2 x^2$ *is diagonal in this basis. What are the diagonal elements?*

Proof. First, recall that in terms of ladder operators, $\hat{x} = \sqrt{\frac{\hbar}{2mw}}(a_+ + a_-)$, and $\hat{p} = i\sqrt{\frac{\hbar m w}{2}}(a_+ - a_-)$.

Also, if $|n'\rangle$ represents the eigenvector of the harmonic oscillator Hamiltonian, then $a_+ |n'\rangle = \sqrt{n'+1}\,|n'+1\rangle$, and $a_- |n'\rangle = \sqrt{n'}\,|n'-1\rangle$ (recall the relation $a_+\psi_n(x) = \sqrt{n+1}\psi_{n+1}(x)$, and $a_-\psi_n(x) = \sqrt{n}\psi_{n-1}(x)$).

Finally, we also have $a_- |0\rangle = 0$ (due to the fact $a_-\psi_0(x) = 0$).

1. Given the above relations, we have the following for all $n \in \mathbb{N}$, and $n' \in \mathbb{Z}_{>0}$:

$$\hat{x}\,|n'\rangle = \sqrt{\frac{\hbar}{2mw}}\left(\sqrt{n'+1}\,|n'+1\rangle + \sqrt{n}\,|n'-1\rangle\right) \implies \langle n|\,\hat{x}\,|n'\rangle = \begin{cases} \sqrt{\frac{\hbar(n'+1)}{2mw}} & n = n'+1 \\ \sqrt{\frac{\hbar n'}{2mw}} & n = n'-1 \\ 0 & \text{otherwise} \end{cases} \tag{53}$$

Which, a special case happens when $n' = 0$, where $\hat{x}\,|0\rangle = \sqrt{\frac{\hbar}{2mw}}\,|1\rangle$, so $\langle n|\,\hat{x}\,|0\rangle = \begin{cases} \sqrt{\frac{\hbar}{2mw}} & n = 1 \\ 0 & \text{otherwise} \end{cases}$.

2. We also have the following for all $n \in \mathbb{N}$ and $n' \in \mathbb{Z}_{>0}$:

$$\hat{p}\,|n'\rangle = i\sqrt{\frac{\hbar m w}{2}}(\sqrt{n'+1}\,|n'+1\rangle - \sqrt{n'}\,|n'-1\rangle) \implies \langle n|\,\hat{p}\,|n'\rangle = \begin{cases} i\sqrt{\frac{\hbar m w(n'+1)}{2}} & n = n'+1 \\ -i\sqrt{\frac{\hbar m w n'}{2}} & n = n'-1 \\ 0 & \text{otherwise} \end{cases} \tag{54}$$

Again, a special case happens when $n' = 0$, where $\hat{p}\,|0\rangle = i\sqrt{\frac{\hbar m w}{2}}\,|1\rangle$, so $\langle n|\,\hat{p}\,|0\rangle = \begin{cases} i\sqrt{\frac{\hbar m w}{2}} & n = 1 \\ 0 & \text{otherwise} \end{cases}$.

3. Based on the results in part 1, if $\mathcal{M}(\hat{x}) = (a_{ij})_{0\le i,j}$ represents the matrix under this basis, then one has:

$$a_{ij} = \begin{cases} 0 & i = j \text{ or } |i-j| \geq 2 \\ \sqrt{\frac{\hbar(j+1)}{2mw}} & i = j+1 \\ \sqrt{\frac{\hbar j}{2mw}} & i = j-1 \end{cases} \tag{55}$$

Writing out the first several rows and columns, we have:

$$\sqrt{\frac{\hbar}{2mw}}\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ 1 & 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \sqrt{4} & \cdots \\ 0 & 0 & 0 & \sqrt{4} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \tag{56}$$

4. Based on th results in part 2, if $\mathcal{M}(\hat{p}) = (b_{jk})_{0\leq j,k}$ represents the matrix under this basis, then one has:

$$b_{jk} = \begin{cases} 0 & j = k \text{ or } |j-k| \geq 2 \\ i\sqrt{\frac{\hbar mw(k+1)}{2}} & j = k+1 \\ -i\sqrt{\frac{\hbar mwk}{2}} & j = k-1 \end{cases} \tag{57}$$

Writing out the first several rows and columns, we have:

$$i\sqrt{\frac{\hbar mw}{2}}\begin{pmatrix} 0 & -1 & 0 & 0 & 0 & \cdots \\ 1 & 0 & -\sqrt{2} & 0 & 0 & \cdots \\ 0 & \sqrt{2} & 0 & -\sqrt{3} & 0 & \cdots \\ 0 & 0 & \sqrt{3} & 0 & -\sqrt{4} & \cdots \\ 0 & 0 & 0 & \sqrt{4} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \tag{58}$$

5. To show $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}mw^2\hat{x}^2$ is diagonal in this basis, we first yield the following computation for all $n' \geq 2$:

$$\hat{p}^2\left|n'\right\rangle = \hat{p}i\sqrt{\frac{\hbar mw}{2}}(\sqrt{n'+1}\left|n'+1\right\rangle - \sqrt{n'}\left|n'-1\right\rangle) \tag{59}$$

$$= -\frac{\hbar mw}{2}(\sqrt{(n'+1)(n'+2)}\left|n'+2\right\rangle - (n'+1)\left|n'\right\rangle - n'\left|n'\right\rangle + \sqrt{n'(n'-1)}\left|n'-2\right\rangle) \tag{60}$$

$$= -\frac{\hbar mw}{2}(\sqrt{(n'+1)(n'+2)}\left|n'+2\right\rangle + \sqrt{n'(n'-1)}\left|n'-2\right\rangle - (2n'+1)\left|n'\right\rangle) \tag{61}$$

$$\hat{x}^2\left|n'\right\rangle = \hat{x}\sqrt{\frac{\hbar}{2mw}}(\sqrt{n'+1}\left|n'+1\right\rangle + \sqrt{n'}\left|n'-1\right\rangle) \tag{62}$$

$$= \frac{\hbar}{2mw}(\sqrt{(n'+1)(n'+2)}\left|n'+2\right\rangle + (n'+1)\left|n'\right\rangle + n'\left|n'\right\rangle + \sqrt{n'(n'-1)}\left|n'-2\right\rangle) \tag{63}$$

$$= \frac{\hbar}{2mw}(\sqrt{(n'+1)(n'+2)}\left|n'+2\right\rangle + (2n'+1)\left|n'\right\rangle + \sqrt{n'(n'-1)}\left|n'-2\right\rangle) \tag{64}$$

Which, the action of $\hat{H}$ is as follow:

$$\begin{align}
\hat{H}\left|n'\right\rangle =& \frac{1}{2m}\hat{p}^2\left|n'\right\rangle + \frac{1}{2}mw^2\hat{x}^2\left|n'\right\rangle \tag{65}\\
=& -\frac{\hbar w}{4}\left(\sqrt{(n'+1)(n'+2)}\left|n'+2\right\rangle + \sqrt{n'(n'-1)}\left|n'-2\right\rangle - (2n'+1)\left|n'\right\rangle\right) \tag{66}\\
&+ \frac{\hbar w}{4}\left(\sqrt{(n'+1)(n'+2)}\left|n'+2\right\rangle + \sqrt{n'(n'-1)}\left|n'-2\right\rangle + (2n'+1)\left|n'\right\rangle\right) \tag{67}\\
=& \frac{\hbar w}{2}(2n'+1)\left|n'\right\rangle = \hbar w\left(n'+\frac{1}{2}\right)\left|n'\right\rangle \tag{68}
\end{align}$$

For special case $n = 0$, one has:

$$\begin{align}
\hat{p}^2\left|0\right\rangle &= \hat{p}i\sqrt{\frac{\hbar mw}{2}}\left|1\right\rangle = -\frac{\hbar mw}{2}(\sqrt{2}\left|2\right\rangle - \left|0\right\rangle) \tag{69}\\
\hat{x}^2\left|0\right\rangle &= \hat{x}\sqrt{\frac{\hbar}{2mw}}\left|1\right\rangle = \frac{\hbar}{2mw}(\sqrt{2}\left|2\right\rangle + \left|0\right\rangle) \tag{70}\\
\implies \hat{H}\left|0\right\rangle &= \frac{1}{2m}\hat{p}^2\left|0\right\rangle + \frac{1}{2}mw^2\hat{x}^2\left|0\right\rangle = -\frac{\hbar w}{4}(\sqrt{2}\left|2\right\rangle - \left|0\right\rangle) + \frac{\hbar w}{4}(\sqrt{2}\left|2\right\rangle + \left|0\right\rangle) = \frac{\hbar w}{2}\left|0\right\rangle \tag{71}
\end{align}$$

Also, for special case $n = 1$, one has:

$$\begin{align}
\hat{p}^2\left|1\right\rangle &= \hat{p}i\sqrt{\frac{\hbar mw}{2}}(\sqrt{2}\left|2\right\rangle - \left|0\right\rangle) = -\frac{\hbar mw}{2}(\sqrt{6}\left|3\right\rangle - 3\left|1\right\rangle) \tag{72}\\
\hat{x}^2\left|1\right\rangle &= \hat{x}\sqrt{\frac{\hbar}{2mw}}(\sqrt{2}\left|2\right\rangle + \left|0\right\rangle) = \frac{\hbar}{2mw}(\sqrt{6}\left|3\right\rangle + 3\left|1\right\rangle) \tag{73}\\
\implies \hat{H}\left|1\right\rangle &= \frac{1}{2m}\hat{p}^2\left|1\right\rangle + \frac{1}{2}mw^2\hat{x}^2\left|1\right\rangle = -\frac{\hbar w}{4}(\sqrt{6}\left|3\right\rangle - 3\left|1\right\rangle) + \frac{\hbar w}{4}(\sqrt{6}\left|3\right\rangle + 3\left|1\right\rangle) = \frac{3\hbar w}{2}\left|1\right\rangle \tag{74}
\end{align}$$

Which, one can see the action of $\hat{H}$ on $\left|n'\right\rangle$ is by scaling with factor $E_n = \hbar w\left(n' + \frac{1}{2}\right)$, which is exactly the eigenvalue / energy of the harmonic oscillator statopmaru states.

□

9

Question 9. *9 A two-level system*
The Hamiltonian in basis $|1\rangle, |2\rangle$ is

$$\hat{H} = \varepsilon(-|1\rangle\langle 1| + |2\rangle\langle 2| + |1\rangle\langle 2| + |2\rangle\langle 1|)$$

1. Find the eigenvalues and eigenvectors.
2. Write the matrix representation of $\hat{H}$ in this basis.

Proof. If input $|1\rangle, |2\rangle$ respectively, we have $\hat{H}|1\rangle = -\epsilon|1\rangle + \epsilon|2\rangle$, and $\hat{H}|2\rangle = \epsilon|1\rangle + \epsilon|2\rangle$ (based on the orthonormality $\langle 1|1\rangle = \langle 2|2\rangle = 1$, and $\langle 1|2\rangle = \langle 2|1\rangle = 0$). Here, we'll assume $\epsilon \neq 0$.

1. Suppose $\lambda \in \mathbb{C}$ is an eigenvalue of $\hat{H}$ and vector $a|1\rangle + b|2\rangle$ is an eigenvector corresponding to it. Then, we have:

$$\lambda(a|1\rangle + b|2\rangle) = \hat{H}(a|1\rangle + b|2\rangle) = a(-\epsilon|1\rangle + \epsilon|2\rangle) + b(\epsilon|1\rangle + \epsilon|2\rangle) = \epsilon(b-a)|1\rangle + \epsilon(b+a)|2\rangle \tag{75}$$

So, $\lambda a = \epsilon(b-a)$, and $\lambda b = \epsilon(b+a)$.

First, suppose $a \neq 0$, WLOG can assume $a = 1$ up to scaling, then $\lambda = \epsilon(b-1)$, plugin to the second equation provides $\epsilon(b-1)b = \epsilon(b+1)$, or $b^2 - 2b - 1 = 0$. So, $b = 1 \pm \sqrt{2}$. Which, these provide $\lambda_\pm = \epsilon(b-1) = \pm\epsilon\sqrt{2}$, each corresponds to eigenvectors $|\lambda_\pm\rangle = |1\rangle + (1 \pm \sqrt{2})|2\rangle$ (up to nonzero scaling).

2. Under ordered basis $\alpha = \{|1\rangle, |2\rangle\}$, $\hat{H}$ is given as:

$$\mathcal{M}(\hat{H}, \alpha) = \epsilon \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \tag{76}$$

If written in the ordered eigenvector basis $\beta = \{|\lambda_+\rangle, |\lambda_-\rangle\}$, then $\hat{H}$ is represented as:

$$\mathcal{M}(\hat{H}, \beta) = \begin{pmatrix} \epsilon\sqrt{2} & 0 \\ 0 & -\epsilon\sqrt{2} \end{pmatrix} \tag{77}$$

□

10

Question 10. *10 A three-level system (Griffiths 3.44)*

$$H = \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix}$$

1. *If the initial state is* $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$*, find* $|S(t)\rangle$.
2. *If the initial state is* $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$*, find* $|S(t)\rangle$.

Proof. Given the hamiltonian as above, given the state $|S(t)\rangle = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}$, the time-dependent Schrödinger equation states:

$$\begin{pmatrix} x'(t) \\ y'(t) \\ z'(t) \end{pmatrix} = \frac{\partial}{\partial t} |S(t)\rangle = \frac{1}{i\hbar} \hat{H} |S(t)\rangle = \begin{pmatrix} a/(i\hbar) & 0 & b/(i\hbar) \\ 0 & c/(i\hbar) & 0 \\ b/(i\hbar) & 0 & a/(i\hbar) \end{pmatrix} \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} \tag{78}$$

Which, compute the characeristic polynomial of the matrix $\frac{1}{i\hbar}\hat{H}$, one has:

$$\det\Big(\hat{H}/(i\hbar) - \lambda I\Big) = \left(\frac{c}{i\hbar} - \lambda\right)\left(\left(\frac{a}{i\hbar} - \lambda\right)^2 - \left(\frac{b}{i\hbar}\right)^2\right) \tag{79}$$

$$= \left(\frac{c}{i\hbar} - \lambda\right)\left(\lambda^2 - \frac{2a}{i\hbar}\lambda + \frac{b^2 - a^2}{\hbar^2}\right) \tag{80}$$

Which, the roots of the characteristic polynomials (or, the eigenvalues) includes $\lambda_1 = \frac{c}{i\hbar}$, and the other two are $\lambda_{2\pm} = \frac{2a/(i\hbar) \pm \sqrt{-4a^2/\hbar^2 + 4(a^2 - b^2)/\hbar^2}}{2} = \frac{a \pm b}{i\hbar}$.

Here, for simplicity assume $a, b, c \in \mathbb{R}$ satisfy $(a+b), (a-b), c$ are distinct (so the three eigenvalues are distinct). Then, the eigenvectors corresponding to each eigenvalue are:

$$\lambda_1 : \quad (\hat{H}/(i\hbar) - \lambda_1 I)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \implies \frac{1}{i\hbar}\begin{pmatrix} a-c & 0 & b \\ 0 & 0 & 0 \\ b & 0 & a-c \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \implies \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \tag{81}$$

$$\implies x, z = 0 \implies \begin{pmatrix} x \\ y \\ z \end{pmatrix} = y \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \tag{82}$$

$$\lambda_{2+}: \quad (\hat{H}/(i\hbar) - \lambda_{2+}I)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \implies \frac{1}{i\hbar}\begin{pmatrix} -b & 0 & b \\ 0 & c-(a+b) & 0 \\ b & 0 & -b \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \implies \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \tag{83}$$

$$\implies y = 0,\ x = z \implies \begin{pmatrix} x \\ y \\ z \end{pmatrix} = z\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \tag{84}$$

$$\lambda_{2-}: \quad (\hat{H}/(i\hbar) - \lambda_{2-}I)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 = implies\frac{1}{i\hbar}\begin{pmatrix} b & 0 & b \\ 0 & c-(a-b) & 0 \\ b & 0 & b \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \implies \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \tag{85}$$

$$\implies y = 0,\ x = -z \implies \begin{pmatrix} x \\ y \\ z \end{pmatrix} = z\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \tag{86}$$

Hence, the general solution to the given Schrödinger equation is spanned by the following three eigenvectors:

$$|\lambda_1\rangle = e^{\lambda_1 t}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = e^{-\frac{ict}{\hbar}}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |\lambda_{2+}\rangle = e^{\lambda_{2+} t}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = e^{-\frac{i(a+b)t}{\hbar}}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad |\lambda_{2-}\rangle = e^{\lambda_{2-} t}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = e^{-\frac{i(a-b)t}{\hbar}}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \tag{87}$$

Which, any state $|S(t)\rangle = a_1 |\lambda_1\rangle + a_{2+} |\lambda_{2+}\rangle + a_{2-} |\lambda_{2-}\rangle$ for some coefficients in $\mathbb{C}$.

1. If the initial state is $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, plugin $t = 0$, one yields $|S(0)\rangle = \begin{pmatrix} a_{2+} - a_{2-} \\ a_1 \\ a_{2+} + a_{2-} \end{pmatrix}$, which enforces $a_1 = 1$, $a_{2+} - a_{2-} = 0$ (or $a_{2+} = a_{2-}$), and $a_{2+} + a_{2-} = 0$ (or $a_{2+} = -a - 2-$). This enforces $a_{2+} = a_{2-} = 0$. So, $|S(t)\rangle = |\lambda_1\rangle$, which is as follow:

$$|S(t)\rangle = e^{-\frac{ict}{\hbar}}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \tag{88}$$

2. If the initial state is $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, we again have $|S(0)\rangle = \begin{pmatrix} a_{2+} - a_{2-} \\ a_1 \\ a_{2+} + a_{2-} \end{pmatrix}$, which enforces $a_1 = 0$, $a_{2+} + a_{2-} = 0$ (or $a_{2-} = -a_{2+}$), and $a_{2+} - a_{2-} = 1$ (which enforces $2a_{2+} = 1$, or $a_{2+} = \frac{1}{2}$, and $a_{2-} = -\frac{1}{2}$). Hence, $|S(t)\rangle = \frac{1}{2}|\lambda_{2+}\rangle - \frac{1}{2}|\lambda_-\rangle$, which is as follow:

$$|S(t)\rangle = \frac{1}{2}e^{-\frac{i(a+b)t}{\hbar}}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2}e^{-\frac{i(a-b)t}{\hbar}}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = e^{-\frac{iat}{\hbar}}\begin{pmatrix} \frac{e^{ibt/\hbar} + e^{-ibt/\hbar}}{2} \\ 0 \\ -\frac{e^{ibt/\hbar} - e^{-ibt/\hbar}}{2} \end{pmatrix} = e^{-iat/\hbar}\begin{pmatrix} \cos(bt/\hbar) \\ 0 \\ -i\sin(bt/\hbar) \end{pmatrix} \tag{89}$$

$\square$